access may require additional approval from a peer or manager, or be constrained by time-bound policies.

These types of dynamic situations require flexible authorization systems that can dynamically respond to requests. Policies can take context (like time or location) into account and respond appropriately or require manager or team-member approval, dynamically granting access when the specified conditions are met.

1.2.3 Maintainability

Resources, roles, and architectures are constantly changing. Having humans or rule-based systems manage and maintain these constantly changing lists leads to over-permissioning which has security implications or under-permissioning, which impacts productivity.

For example, imagine a SaaS project management platform that allows customers to create projects and gives employees different levels of access to tasks, files, and reports. We might allow project owners to manage everything, team members to view and modify tasks assigned to them, external contractors to view only tasks assigned to them, and finance team members to access billing reports but not project details.

Whenever a new task is created, a project changes owners or team membership changes, ACLs must be updated. If a contractor’s role changes multiple lists might be affected. If a contractor needs temporary edit access, you might have to create a new role called Contractor-Editor. Cross-team access results in role proliferation creating numerous special-purpose roles for unique issues. Role proliferation complicates management and can obscure the impact of any single role assignment, making it difficult to predict or audit authorization decisions—especially when principals belong to dozens or even hundreds of overlapping roles.

In contrast, dynamic authorization accounts for these kinds of changes and exceptions by evaluating access permission at runtime. When roles or assignments change, access adjusts automatically.